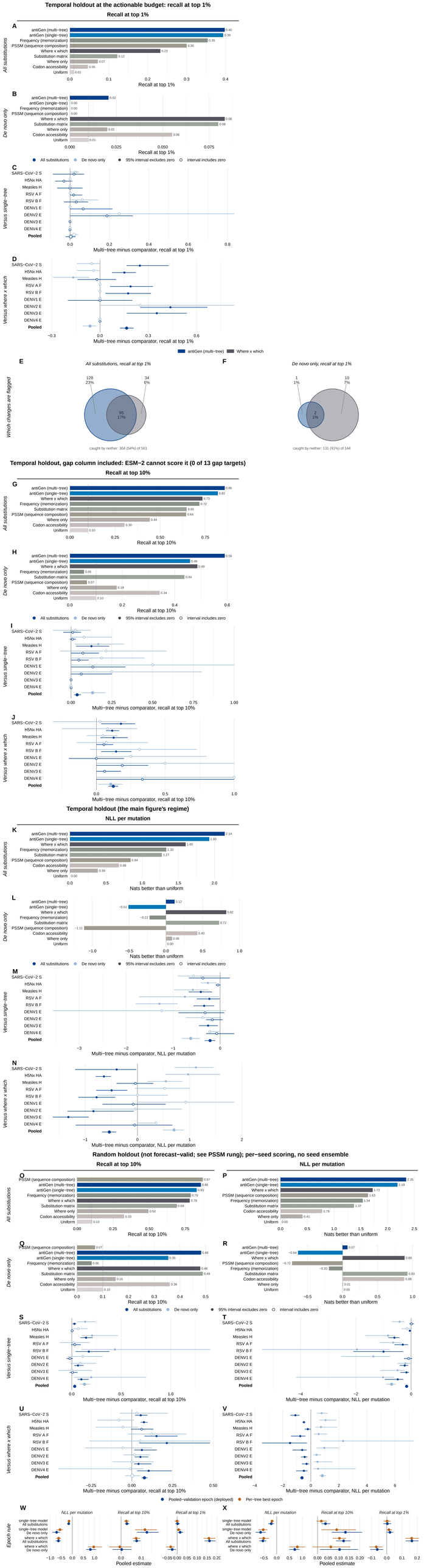
Temporal holdout at the actionable budget: recall at top 1%
Recall at top 1%
All substitutions
De novo only
Versus single−tree
Versus where x which
antiGen (multi−tree)
antiGen (single−tree)
Frequency (memorization)
PSSM (sequence composition)
Where x which
Substitution matrix
Where only
Codon accessibility
Uniform
0.40
0.39
0.35
0.30
0.23
0.12
0.07
0.05
0.01
0.02
0.00
0.00
0.00
0.08
0.08
0.02
0.06
0.01
Recall at top 1%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
Multi−tree minus comparator, recall at top 1%
Multi−tree minus comparator, recall at top 1%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 1%
128
23%
34
6%
95
17%
caught by neither: 304 (54%) of 561
F
De novo only, recall at top 1%
1
1%
10
7%
2
1%
caught by neither: 131 (91%) of 144
Which changes are flagged
Temporal holdout, gap column included: ESM−2 cannot score it (0 of 13 gap targets)
Recall at top 10%
0.86
0.82
0.73
0.72
0.65
0.64
0.44
0.30
0.10
0.59
0.46
0.49
0.05
0.44
0.07
0.18
0.34
0.10
Recall at top 10%
Multi−tree minus comparator, recall at top 10%
Multi−tree minus comparator, recall at top 10%
Temporal holdout (the main figure's regime)
NLL per mutation
2.14
1.93
1.60
1.33
1.27
0.84
0.68
0.39
0.00
Nats better than uniform
0.12
−0.51
0.82
−0.22
0.72
−1.11
0.43
0.08
0.00
Nats better than uniform
Multi−tree minus comparator, NLL per mutation
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Recall at top 10%
NLL per mutation
PSSM (sequence composition)
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
Where only
Codon accessibility
Uniform
0.87
0.86
0.83
0.79
0.76
0.69
0.50
0.33
0.10
2.35
2.19
1.72
1.63
1.54
1.37
0.78
0.41
0.00
Recall at top 10%
Nats better than uniform
0.07
0.48
0.35
0.06
0.48
0.49
0.15
0.36
0.10
−0.64
0.69
−0.72
−0.20
0.93
0.88
0.01
0.00
Recall at top 10%
Nats better than uniform
Multi−tree minus comparator, recall at top 10%
Multi−tree minus comparator, NLL per mutation
Multi−tree minus comparator, recall at top 10%
Multi−tree minus comparator, NLL per mutation
Pooled−validation epoch (deployed)
Per−tree best epoch
Epoch rule
W
X
NLL per mutation
Recall at top 10%
Recall at top 1%
single−tree model
All substitutions
De novo only
where x which
Pooled estimate